

Online sampled-budget branch outcomes  
Independent process per policy; paired conclusions require the separate prefix audit

Continuous straight

Continuous recenter + align

aligned\_control | d=18 mm,  $\mu=1/1$ , a=0 deg, B=4 N

Clear / safe

Clear / safe

angle\_control | d=18 mm,  $\mu=1/1$ , a=1 deg, B=4 N

Clear / safe

Clear / safe

candidate | d=18 mm,  $\mu=1/1$ , a=1.5 deg, B=4 N

Recovery budget exceeded

Clear / safe

higher\_budget | d=18 mm,  $\mu=1/1$ , a=1.5 deg, B=5 N

Clear / safe

Clear / safe

shallower\_depth | d=12 mm,  $\mu=1/1$ , a=1.5 deg, B=4 N

Clear / safe

Clear / safe

lower\_friction | d=18 mm,  $\mu=0.5/0.5$ , a=1.5 deg, B=4 N

Reference budget exceeded

not tested

grasp\_risk | d=18 mm,  $\mu=1/1$ , a=2 deg, B=4 N

Reference budget exceeded

not tested

lower\_budget | d=18 mm,  $\mu=1/1$ , a=1.5 deg, B=3 N

Reference budget exceeded

not tested